Atulya Mohan

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Education

Carnegie Mellon University – MS in Engineering & Technology Innovation Management — Jan 2025 – Dec 2025

Coursework: Electromechanical Systems Design, Data Science, Quantitative Entrepreneurship, Product Management

Purdue University – BS in Mechanical Engineering — Aug 2020 – May 2024

Coursework: Machine Design, Controls, FEA, Statics & Dynamics, Fluid Mechanics, Thermodynamics, Heat & Mass Transfer

Experience

RoCo Global, NPI Engineer Intern – Pittsburgh, PA — Jan 2026 – Present

- Designed and prototyped a custom automated sieve-feeding mechanism in Fusion 360 for clump breakdown of proprietary PU foam powder, increasing manufacturing throughput by 300%
- Coordinating with three external laboratories to test PU foam-additive cementitious and spray-foam formulations, validating physical properties against applicable ASTM standards

Fourier, Hardware Engineering Intern – Mountain View, CA — May 2025 – Aug 2025

- Inherited the hydrogen dryer program with zero internal documentation after the lead engineer's departure; conducted temperature-profile and time-to-saturation characterization experiments on the existing PSA dryer, determining that in-situ TSA conversion was infeasible under operating conditions
- Proposed and built a 3-column dryer architecture using aluminum columns with ex-situ desiccant regeneration to meet a 1.5-month pilot deadline; selected aluminum over stainless steel to reduce fabrication lead time and evaluated molecular sieve versus activated alumina for lower-heat regeneration
- Validated the pilot-ready dryer on schedule, extending continuous drying time from 4 hours to approximately 30 hours (3.5 days between desiccant replacements); confirmed ex-situ regeneration feasibility, minimizing desiccant waste
- Authored a comprehensive knowledge transfer document detailing characterization data, design rationale, and recommendations to guide future in-situ regeneration dryer development
- Designed and built a sheet-metal test stand to evaluate power-supply behavior and troubleshoot I2C communication issues

Michelin, Research & Development Intern – Pune, India — Sep 2024 – Dec 2024

- Analyzed lab-measured mold geometry, mapped mold codes to production tires and irregular-wear reports, and quantified the contribution of mold geometry to final tire-geometry variation

Purdue University (BIPL), Undergraduate Research Assistant – West Lafayette, IN — May 2023 – Dec 2023

- Designed and simulated strain field behavior of 2D dielectric elastomer actuator arrays in Abaqus by mapping temperature to applied voltage with equibiaxial prestretch and thermally coupled elements, leading to a co-authored publication

Michelin North America (PRIME), Process Quality Engineering Intern – Greenville, SC — May 2022 – Aug 2022

- Investigated a high-priority scrap defect by inspecting 500+ tires and recording multi-parameter data; designed a go/no-go gauge, iterated 3D-printed prototypes with operator feedback, and drove a 100-unit stainless steel rollout using DFM principles

Projects

Launch-A-Birdie: Automatic badminton shuttlecock launcher — Aug 2025 – Dec 2025

- Designed and assembled a robotic launcher in Fusion 360, integrating feeding, actuation, and aiming subsystems; wrote Arduino firmware coordinating servo and stepper mechanisms and validated full-system operation through iterative prototype builds

Custom PC Case: Sheet metal chassis design & prototyping — Sep 2025

- Designed a chassis optimized for sheet metal fabrication in Fusion 360 with attention to bend allowances, fastener integration, and internal component clearances; validated fitment via full-scale 3D-printed prototypes applying DFM principles throughout

Navigation Device: Wearable assistive necklace for the visually impaired — Jan 2024 – May 2024

- Led technical design for a wearable necklace integrating 8 ultrasonic sensors, 8 infrared sensors, and 8 haptic motors to relay 360° environmental information; implemented I2C protocol with an Arduino Nano to coordinate all peripherals, reducing device size and wiring complexity

Technical Skills

Design & Simulation Tools: SOLIDWORKS | Siemens NX | Fusion 360 | Abaqus | COMSOL | Figma

Programming & Data Tools: MATLAB | Python | Arduino (C++) | R | Excel | Power BI